Health records and EEG reports: TUAB, the TUH training cohort and HEEDB

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What the three corpora we train on say about their patients, beyond the EEG: **age and sex**, the **normal / abnormal** verdict of the EEG report, the patients' **pathologies** (seizure disorders, stroke, dementia, psychiatric diagnoses, headache and migraine, sleep disorders, tumours and others) and the **types of abnormality** the reports name (slowing, epileptiform discharges, seizures, suppression, ...). Every statistic is computed on one harmonised per-recording table (`labram/eval/clinical_analytics.py`), so the corpora read side by side.

- **TUAB** (v3.0.1, 2,993 recordings): the normal / abnormal benchmark; its test split is our TUAB eval set.
- **TUH cohort** (`all_both_v3_anymiss` , 31,312 recordings): the TUH part of E2 / E2L / E2P.
- **HEEDB** (`heedb_routine_v1` , 71,805 adult routine recordings): the Harvard part of the same runs.

HEEDB is under a data use agreement: every HEEDB cell computed from fewer than 10 recordings is blank, and no row-level data is shown.

1. Sources and definitions

Table 1.1. Where each kind of information comes from

	TUAB / TUH cohort	HEEDB
Age, sex	EDF header (age), cohort metadata	BDSP metadata (`AgeAtVisit` , sex)
Normal / abnormal	TUAB folder label; else the NEDC report finding (TUH cohort)	the EEG report (BDSP findings), unlabelled when the session has no report
Pathologies	NEDC TUEG clinical sheet: 15 Y/N diagnosis columns per session, plus keyword mentions in its free-text "Other diagnosis"	per-patient ICD-10 codes (neurology extract): TUH-aligned `dx_*` flags and ICD-10 groups `icd_*`
Abnormality types	the sheet's "abnormal due to" text, mapped to reason categories (`meta_analysis.REPORT_REASONS`)	report finding groups (`fg_*`)
Missing information	session not in the sheet → unknown	patient without any ICD-10 record → unknown

Three caveats shape every comparison below:

- **HEEDB pathologies are per patient and undated.** An ICD-10 code may predate or postdate the EEG; the TUH sheet describes the session. HEEDB rates are therefore "ever diagnosed", TUH rates "noted at this EEG".
- **TUH free-text categories are mentions.** Headache / migraine, sleep disorder, tumour, syncope and the other keyword categories (Table 1.2, marked *mention*) count a session when its "Other diagnosis" text names the condition; a session that does not name it is counted as not mentioned, which is not evidence of absence. These rates are **lower bounds**.

- **The NEDC sheet leaves its diagnosis columns almost empty for TUAB sessions** (seizure 0.1 %, epilepsy 0.04 % against 8.4 % / 3.4 % for the other TUEG sessions of the cohort), although 50 % of them have "Other diagnosis" text and 35 % an "abnormal due to" reason. TUAB's near-zero diagnosis rates mean *not annotated*, not healthy.

A corpus that does not code a category at all (e.g. HEEDB's neurology extract has no psychiatric or injury codes besides depression and ADHD) is left blank, never 0.

Table 1.2. Pathology categories and their source columns

`tuh` / `heedb` list the columns OR-ed into the category (`kw_*` = keyword mention in the TUH free text, `dx_*` / `icd_*` = coded diagnosis); `–` = not coded in that corpus.

category	tuh	heedb	TUH kind
seizure_epilepsy	dx_clinical_seizure, dx_epilepsy, dx_electrographic_seizure	dx_clinical_seizure, dx_epilepsy	Y/N column
stroke_cerebrovascular	dx_stroke	dx_stroke, icd_cerebrovascular_diseases	Y/N column
dementia	dx_dementia, dx_alzheimers	dx_dementia, dx_alzheimers	Y/N column
parkinson_movement	dx_parkinsons	dx_parkinsons, icd_movement_and_cerebellar_disorders	Y/N column
multiple_sclerosis	dx_ms	dx_ms	Y/N column
tbi	dx_tbi	–	Y/N column
depression	dx_depression	dx_depression	Y/N column
anxiety	dx_anxiety	–	Y/N column
bipolar	dx_bipolar	–	Y/N column
schizophrenia	dx_schizophrenia	–	Y/N column
substance_use	dx_substance	–	Y/N column
adhd	dx_adhd	dx_adhd	Y/N column
headache_migraine	kw_headache_migraine	icd_headache_disorders	mention (lower bound)
sleep_disorder	kw_sleep_disorder	icd_sleep_disorders	mention (lower bound)
brain_tumor	kw_brain_tumor	icd_intracranial_and_spinal_tumors	mention (lower bound)
developmental	kw_developmental	icd_developmental_delay	mention (lower bound)
cns_infection	kw_cns_infection	icd_infections	mention (lower bound)
syncope	kw_syncope	–	mention (lower bound)
altered_mental_status	kw_altered_mental_status	–	mention (lower bound)
anoxic_injury	kw_anoxic_injury	–	mention (lower bound)
vascular_risk	kw_vascular_risk	–	mention (lower bound)

Table 1.3. Recordings with the pathology known, and positives

Per corpus and category: `known` = recordings whose status for the category is known (TUH: the session is in the sheet; HEEDB: the patient has an ICD-10 record), `positive` = those with the category. HEEDB counts under 10 are blank.

pathology	TUAB	TUH cohort	known HEEDB	TUAB	TUH cohort	positive HEEDB
seizure_epilepsy	2559	28554	65020	4	2573	23035
stroke_cerebrovascular	2559	28554	65020	6	1122	9466
dementia	2559	28554	65020	1	83	4990
parkinson_movement	2559	28554	65020	0	4	9477
multiple_sclerosis	2559	28554	65020	10	66	934
tbi	2559	28554	0	2	364	0
depression	2559	28554	65020	3	129	13024
anxiety	2559	28554	0	0	122	0
bipolar	2559	28554	0	42	321	0
schizophrenia	2559	28554	0	0	92	0
substance_use	2559	28554	0	106	1214	0
adhd	2559	28554	65020	3	35	1843
headache_migraine	2559	28554	65020	135	617	17954
sleep_disorder	2559	28554	65020	28	239	14356
brain_tumor	2559	28554	65020	51	677	4126
developmental	2559	28554	65020	34	419	1878
cns_infection	2559	28554	65020	31	537	1781
syncope	2559	28554	0	185	1176	0
altered_mental_status	2559	28554	0	166	2326	0
anoxic_injury	2559	28554	0	25	931	0
vascular_risk	2559	28554	0	384	4026	0

2. Who the recordings come from: age, sex and the EEG verdict

Table 2.1. Corpus overview

Recordings, distinct patients, age (mean, SD, years), the share of women, the EEG report's verdict (normal / abnormal / unknown, % of recordings) and the share with health-record information (Table 1.1).

corpus	recordings	patients	age mean	age SD	female %	normal %	abnormal %	status unknown %	health record %
TUAB	2993	2329	49.1	17.4	53.3	50.8	49.2	0.0	85.5
TUH cohort	31312	13978	52.3	17.5	51.1	39.2	53.0	7.9	91.2
HEEDB	71805	57678	54.0	19.8	51.5	20.1	30.6	49.2	90.6

Figure 2.1. Age distribution by EEG verdict

Share of recordings per 5-year age bin (each curve sums to 100 %), for normal (solid) and abnormal (dashed) reports; adults 18+. (a) TUAB and the TUH cohort; (b) HEEDB.

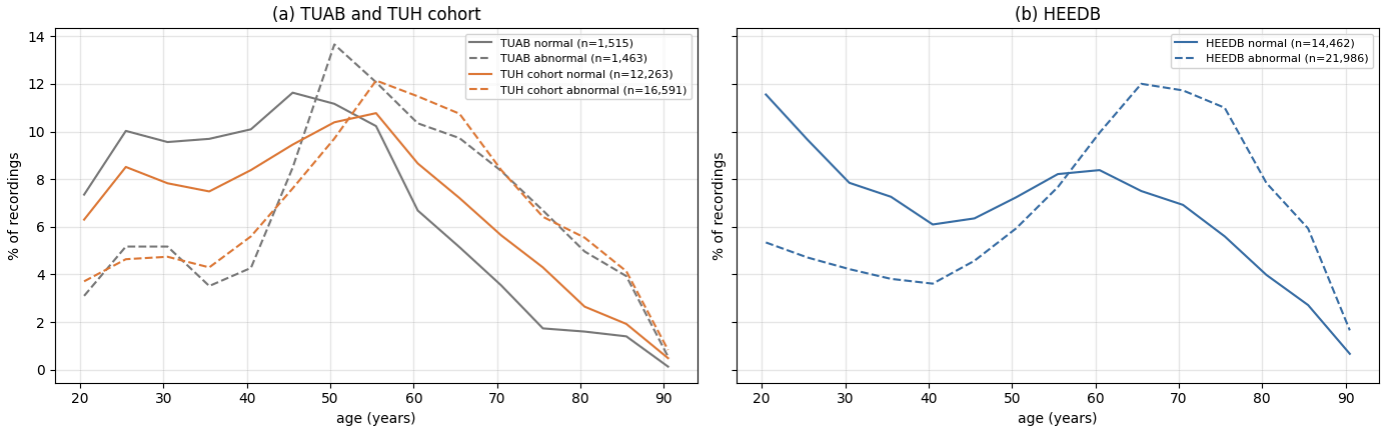


Table 2.2. TUAB: recordings by age group, sex and EEG verdict

Counts per age group (rows) and sex × verdict (columns); recordings with unknown sex are left out.

sex	F		M	
status	abnormal	normal	abnormal	normal
age				
18-29	67	202	80	118
30-39	60	186	61	104
40-49	126	168	124	169
50-59	151	141	187	142
60-69	159	88	141	76
70-79	102	28	84	33
80+	67	25	41	16

Table 2.3. TUH cohort: recordings by age group, sex and EEG verdict

Counts per age group (rows) and sex × verdict (columns); recordings with unknown sex are left out.

sex	F			M		
status	abnormal	normal	unknown	abnormal	normal	unknown
age						
18-29	751	1353	191	957	856	139
30-39	738	1095	134	773	788	114
40-49	1121	1231	187	1301	1041	218
50-59	1786	1255	274	2047	1265	325
60-69	1676	900	224	1857	900	271
70-79	1166	608	155	1038	455	87
80+	802	262	104	553	223	35

Table 2.4. HEEDB: recordings by age group, sex and EEG verdict

Counts per age group (rows) and sex × verdict (columns); recordings with unknown sex are left out. Cells under 10 recordings are blank (DUA).

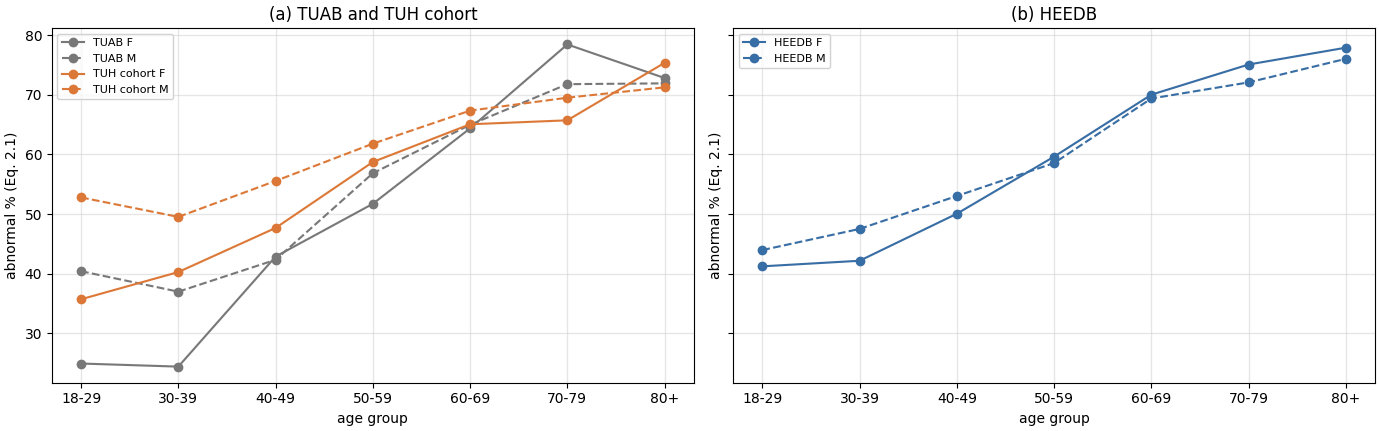
sex			F			M
status	abnormal	normal	unknown	abnormal	normal	unknown
age						
18-29	1303	1858	3255	1300	1658	2703
30-39	849	1165	2308	855	945	1600
40-49	1015	1013	2461	937	830	1921
50-59	1703	1153	3063	1640	1161	2592
60-69	2355	1008	3320	2693	1188	3204
70-79	2243	744	2802	2449	948	2791
80+	1316	373	1708	1327	418	1584

Figure 2.2. Share of abnormal EEGs by age group and sex

Abnormal % among recordings with a known verdict, Eq. (2.1), per age group; women solid, men dashed. (a) TUAB and the TUH cohort; (b) HEEDB. A point needs at least 10 recordings.

$$a_g = 100\,\frac{\#\{i \in g : \text{report}_i = \text{abnormal}\}}{\#\{i \in g : \text{report}_i \in \{\text{normal}, \text{abnormal}\}\}} \tag{2.1}$$

where g is a set of recordings (an age group of one sex and corpus).



3. Pathologies

For a corpus and category p, the prevalence counts only the recordings whose status for p is known (Table 1.3):

$$\pi_p = 100\,\frac{\#\{i : x_{ip} = 1\}}{\#\{i : x_{ip} \in \{0, 1\}\}} \tag{3.1}$$

where $x_{ip} = 1$ when recording i has category p, 0 when it is known not to (or, for a mention category, not mentioned), and unknown otherwise (`clinical_analytics.prevalence`).

Table 3.1. Prevalence per corpus (%)

Eq. (3.1) per category and corpus; *mention* categories (Table 1.2) are lower bounds for TUH; blank = not coded (or a HEEDB count under 10).

	TUAB	TUH cohort	HEEDB
seizure_epilepsy	0.2	9.0	35.4
stroke_cerebrovascular	0.2	3.9	14.6
dementia	0.0	0.3	7.7
parkinson_movement	0.0	0.0	14.6
multiple_sclerosis	0.4	0.2	1.4
tbi	0.1	1.3	–
depression	0.1	0.5	20.0
anxiety	0.0	0.4	–
bipolar	1.6	1.1	–
schizophrenia	0.0	0.3	–
substance_use	4.1	4.3	–
adhd	0.1	0.1	2.8
headache_migraine	5.3	2.2	27.6
sleep_disorder	1.1	0.8	22.1
brain_tumor	2.0	2.4	6.3
developmental	1.3	1.5	2.9
cns_infection	1.2	1.9	2.7
syncope	7.2	4.1	–
altered_mental_status	6.5	8.1	–
anoxic_injury	1.0	3.3	–
vascular_risk	15.0	14.1	–

Figure 3.1. Prevalence by age group

Eq. (3.1) per category (rows) and age group (columns), %. (a) TUH cohort; (b) HEEDB. Categories not coded in the corpus are left out; HEEDB cells under 10 positives are blank.

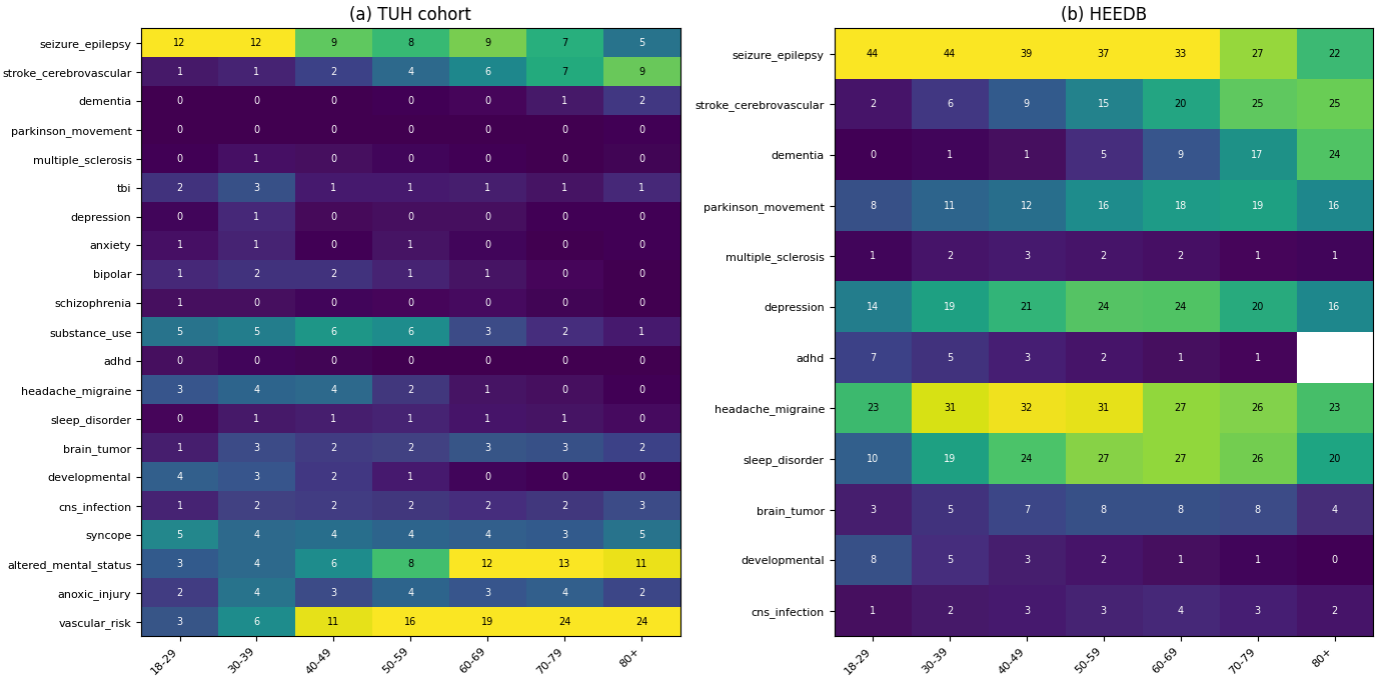


Table 3.2. Prevalence by sex (%)

Eq. (3.1) per corpus and sex.

	TUAB F	TUAB M	TUH cohort F	TUH cohort M	HEEDB F	HEEDB M
seizure_epilepsy	0.2	0.1	8.7	9.3	35.8	35.1
stroke_cerebrovascular	0.3	0.2	3.7	4.2	13.7	15.5
dementia	0.0	0.1	0.3	0.3	7.3	8.1
parkinson_movement	0.0	0.0	0.0	0.0	14.0	15.2
multiple_sclerosis	0.4	0.3	0.4	0.1	1.9	0.9
tbi	0.0	0.2	0.7	1.9	–	–
depression	0.1	0.1	0.6	0.3	22.6	17.3
anxiety	0.0	0.0	0.5	0.4	–	–
bipolar	2.1	1.2	1.4	0.9	–	–
schizophrenia	0.0	0.0	0.1	0.5	–	–
substance_use	2.5	6.0	2.9	5.6	–	–
adhd	0.1	0.1	0.1	0.1	2.6	3.1
headache_migraine	7.4	2.9	3.0	1.3	32.2	22.7
sleep_disorder	1.3	0.8	0.8	0.9	21.1	23.1
brain_tumor	2.2	1.7	2.7	2.0	6.6	6.0
developmental	1.2	1.5	1.2	1.7	2.4	3.4
cns_infection	1.3	1.2	2.0	1.8	2.6	2.9
syncope	6.9	7.6	4.6	3.7	–	–
altered_mental_status	6.1	6.9	7.9	8.4	–	–
anoxic_injury	1.0	1.0	3.3	3.2	–	–
vascular_risk	14.3	15.8	14.0	14.1	–	–

Table 3.3. How often the EEG is abnormal with and without each pathology

Among recordings with a known verdict, the abnormal share, Eq. (2.1), of those with the category (`with`) and of those known not to have it (`without`); categories with at least 10 recordings.

$$\Delta a_p = a_{\{i:\, x_{ip}=1\}} - a_{\{i:\, x_{ip}=0\}} \tag{3.2}$$

is the difference in percentage points (`clinical_analytics.abnormal_rate_by_pathology`).

	abnormal % with			Δ (pp), Eq. 3.2	
corpus	**TUAB**	**TUH cohort**	**HEEDB**	**TUAB**	**TUH cohort**	**HEEDB**
pathology						
seizure_epilepsy	–	60.9	65.9	–	4.0	9.4
stroke_cerebrovascular	–	82.4	70.6	–	26.2	12.8
dementia	–	72.3	74.2	–	15.1	15.4
parkinson_movement	–	–	64.2	–	–	4.8
multiple_sclerosis	60.0	51.5	57.1	16.1	-5.8	-3.3
tbi	–	69.0	–	–	11.8	–
depression	–	44.2	59.9	–	-13.2	-0.5
anxiety	–	48.4	–	–	-9.0	–
bipolar	33.3	58.9	–	-10.8	1.6	–
schizophrenia	–	51.1	–	–	-6.2	–
substance_use	43.4	71.1	–	-0.5	14.4	–
adhd	–	42.9	45.1	–	-14.4	-15.8
headache_migraine	10.4	32.6	60.0	-35.4	-25.2	-0.4
sleep_disorder	39.3	55.2	59.8	-4.7	-2.1	-0.7
brain_tumor	66.7	70.8	78.0	23.2	13.8	19.4
developmental	58.8	60.9	59.6	15.1	3.6	-0.8
cns_infection	64.5	66.1	77.5	20.8	9.0	17.9
syncope	22.7	34.6	–	-22.9	-23.6	–
altered_mental_status	75.9	78.7	–	34.2	23.3	–
anoxic_injury	92.0	80.5	–	48.6	24.0	–
vascular_risk	53.1	65.5	–	10.8	9.6	–

Figure 3.2. Abnormal-EEG excess by pathology

Δa_p (Eq. (3.2), percentage points) per category: (a) TUH cohort; (b) HEEDB. Positive = the EEG is more often abnormal in patients with the category than in those without it.

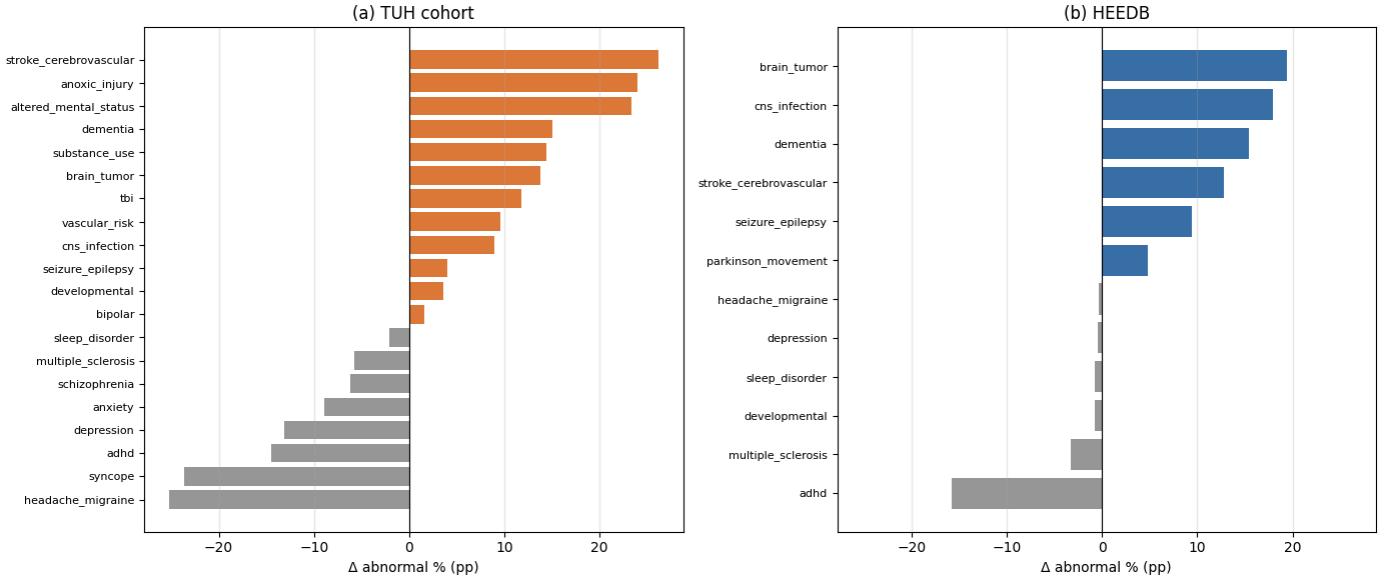


Figure 3.3. Headache / migraine: prevalence by age and sex

Eq. (3.1) for the headache / migraine category per age group, women solid, men dashed: (a) TUAB and the TUH cohort (mentions in the "Other diagnosis" text, lower bounds); (b) HEEDB (ICD-10 headache disorders, G43/G44).

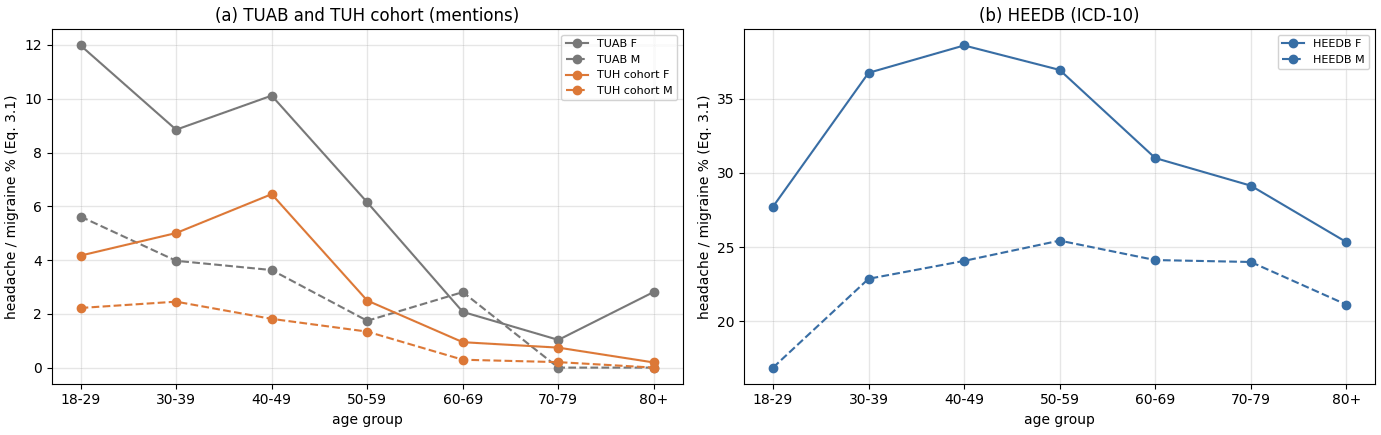


Table 3.4. HEEDB: co-occurrence of pathologies

For each row category, the share (%) of its patients' recordings that also carry the column category:

$$c_{pq} = 100\,\frac{\#\{i : x_{ip} = 1,\ x_{iq} = 1\}}{\#\{i : x_{ip} = 1,\ x_{iq} \in \{0, 1\}\}} \tag{3.3}$$

(`clinical_analytics.co_occurrence`); the diagonal is 100. Cells under 10 recordings are blank.

	seizure_epilepsy	stroke_cerebrovascular	de-mentia	parkinson_movement	multiple_sclerosis	de-pres-sion	adhd	headache_migraine	s
seizure_epilepsy	100	19	9	21	2	29	4	41	
stroke_cerebrovascular	46	100	17	29	2	34	2	53	
dementia	43	33	100	32	2	40	2	39	
parkinson_movement	50	29	17	100	4	42	5	49	
multiple_sclerosis	42	16	9	38	100	36	4	43	
depression	50	25	15	31	3	100	7	52	
adhd	56	11	6	25	2	51	100	48	
headache_migraine	52	28	11	26	2	38	5	100	
sleep_disorder	47	25	14	31	2	44	6	49	
brain_tumor	62	21	8	21	1	28	3	54	
developmental	73	9	6	27	1	28	12	33	
cns_infection	55	29	14	33	5	33	5	54	

Table 3.5. TUH cohort: co-occurrence of pathologies

Eq. (3.3) for the TUH cohort, categories with at least 50 positives.

	seizure_epilepsy	stroke_cerebrovascular	de-mentia	multiple_sclerosis	tbi	de-pres-sion	anx-iety	bipo-lar	schizophre-nia	substance_use	hea
seizure_epilepsy	100	13	1	0	5	1	2	1	2	7	
stroke_cerebrovascular	29	100	3	0	6	1	1	2	2	5	
dementia	33	36	100	0	1	2	2	0	5	2	
multiple_sclerosis	0	0	0	100	5	0	0	5	0	6	
tbi	37	18	0	1	100	1	1	0	0	7	
depression	26	5	2	0	3	100	37	5	3	24	
anxiety	44	12	2	0	3	39	100	8	4	34	
bipolar	9	6	0	1	0	2	3	100	4	10	
schizophrenia	57	23	4	0	0	4	5	13	100	8	
substance_use	15	5	0	0	2	3	3	3	1	100	
headache_migraine	7	1	0	2	1	0	3	2	0	3	
sleep_disorder	10	5	1	0	0	1	1	2	2	2	
brain_tumor	17	1	0	0	0	0	0	0	0	0	
developmental	8	1	0	0	3	0	0	1	0	2	
cns_infection	11	1	0	0	0	0	0	0	0	1	
syncope	3	2	0	1	0	0	0	2	0	3	
altered_mental_status	7	6	1	0	1	0	0	1	1	4	
anoxic_injury	5	1	0	0	1	0	0	1	0	6	
vascular_risk	5	5	0	0	1	1	0	2	1	5	

4. What makes an EEG abnormal: abnormality types

Among abnormal recordings whose report gives a reason (TUH: an "abnormal due to" text) or findings (HEEDB), the share naming abnormality type t:

$$s_t = 100\,\frac{\#\{i \in A : y_{it} = 1\}}{\#A} \tag{4.1}$$

where A is that set of abnormal recordings and y_{it} marks type t in recording i's report (`clinical_analytics.abnormal_types`). A report can name several types, so the shares do not sum to 100. TUH types come from keyword rules on free text, HEEDB types from BDSP's

structured findings; a type HEEDB does not code (posterior rhythm, excess beta, asymmetry) is blank.

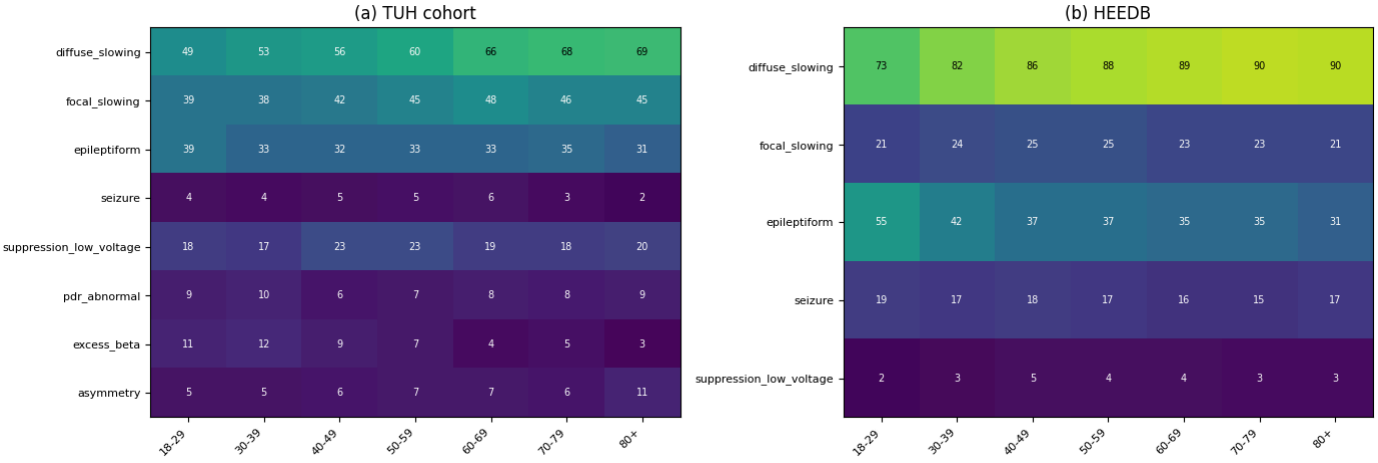
Table 4.1. Abnormality types per corpus (%)

`n` = abnormal recordings with a reason / findings (the denominator of Eq. (4.1)).

	n	diffuse_slowing	focal_slowing	epileptiform	seizure	suppression_low_voltage	pdr_abnormal	excess_beta	asymmetry
TUAB	1024.0	59.4	52.4	37.6	1.8	15.4	7.8	6.8	7.6
TUH cohort	16276.0	60.7	44.0	33.7	4.4	20.2	7.9	6.8	6.8
HEEDB	21986.0	86.5	22.9	37.9	16.5	3.6	–	–	–

Figure 4.1. Abnormality types by age group

Eq. (4.1) per type (rows) and age group (columns), %: (a) TUH cohort; (b) HEEDB.



Tables 4.2–4.3. Which abnormality types go with which pathology

The lift of type t among abnormal recordings with pathology p over all abnormal recordings of the corpus:

$$L_{pt} = \frac{s_t(A \cap \{i : x_{ip} = 1\})}{s_t(A)} \tag{4.2}$$

with s_t of Eq. (4.1) (`clinical_analytics.type_by_pathology_lift`). $L > 1$: the type is over-represented in that pathology (e.g. epileptiform discharges in epilepsy). Pathologies with at least 10 such recordings; HEEDB cells under 10 positives are blank.

Table 4.2. TUH cohort: lift of abnormality types by pathology

	diffuse_slowing	focal_slowing	epileptiform	seizure	suppression_low_voltage	pdr_abnormal	excess_beta	asymmetry
seizure_epilepsy	0.98	1.05	1.27	4.35	0.72	0.94	0.77	0.33
stroke_cerebrovascular	1.13	1.26	1.07	2.10	1.35	0.49	0.54	0.61
dementia	1.40	1.06	1.19	0.75	0.58	1.05	1.23	–
multiple_sclerosis	1.07	1.07	1.05	–	1.45	0.74	1.74	1.74
tbi	1.01	1.32	0.89	1.63	1.07	1.06	0.83	0.12
depression	0.78	0.64	0.52	3.96	0.35	0.44	2.86	–
anxiety	0.59	1.15	0.76	–	0.17	2.14	2.76	–
bipolar	1.08	0.72	1.00	1.33	1.03	1.01	2.77	0.71
schizophrenia	1.19	0.92	0.63	–	1.79	0.54	–	0.63
substance_use	1.07	0.66	0.79	0.58	1.01	1.16	1.93	0.57
adhd	1.43	0.45	2.18	–	0.33	–	0.99	–
headache_migraine	0.77	1.39	0.97	1.59	0.35	0.51	0.45	0.67
sleep_disorder	0.93	0.85	0.59	–	0.72	1.73	0.34	0.34
brain_tumor	1.02	1.72	1.36	1.56	0.63	1.09	0.62	1.46
developmental	1.03	0.76	1.43	0.54	0.81	1.51	2.83	1.06
cns_infection	1.05	1.31	1.44	1.65	0.78	1.07	0.67	1.29
syncope	1.00	0.98	0.70	–	0.61	1.25	0.48	1.10
altered_mental_status	1.27	0.92	0.92	0.83	0.95	0.71	0.80	1.01
anoxic_injury	0.79	0.32	0.85	0.84	2.51	0.66	0.89	0.43
vascular_risk	1.08	0.93	0.75	0.51	0.96	1.39	0.65	1.03

Table 4.3. HEEDB: lift of abnormality types by pathology

	diffuse_slowing	focal_slowing	epileptiform	seizure	suppression_low_voltage
seizure_epilepsy	0.98	1.25	1.28	1.05	0.83
stroke_cerebrovascular	1.04	1.16	0.86	0.78	0.96
dementia	1.04	0.91	0.78	0.77	0.63
parkinson_movement	1.02	0.90	0.90	0.84	1.06
multiple_sclerosis	1.00	0.80	0.96	0.73	0.95
depression	1.02	0.95	0.92	0.82	0.97
adhd	0.95	0.94	1.05	0.90	0.70
headache_migraine	1.02	1.27	0.99	0.85	0.75
sleep_disorder	1.02	0.96	0.91	0.76	0.91
brain_tumor	1.04	1.69	1.03	0.89	0.27
developmental	0.95	0.75	1.31	1.05	0.57
cns_infection	1.05	1.02	1.03	0.92	1.35

5. Findings

Demographics and verdict (Tables 2.1–2.4, Figures 2.1–2.2).

- **Age and sex mix.** The three corpora are adult and close in age (mean 49 years for TUAB, 52 for the TUH cohort, 54 for HEEDB) and about half women (51–53 %).
- **Report coverage.** TUAB is balanced by design (51 % normal). The TUH cohort is 53 % abnormal. Half of the HEEDB recordings have no report verdict (49 % unknown); among the rest, 60 % are abnormal.
- **Abnormal share rises with age** in every corpus (Eq. (2.1), Figure 2.2): from 30–44 % under 40 to 72–77 % at 80+. At every age men are slightly more often abnormal than women (TUAB 52 vs 47 %, TUH cohort 61 vs 55 %, HEEDB 61 vs 60 %).
- **Consequence for training.** Age and abnormality are strongly confounded. A brain-age model trained on all recordings sees mostly abnormal EEGs at old age, and a pathology model can lean on age-related EEG change (cf. the bias-corrected brain-age gap in `docs/brain_age_bias_correction.md`).

Pathologies (Tables 3.1–3.5, Figures 3.1–3.3).

- **The corpora describe pathology very differently.**
  - HEEDB's per-patient ICD-10 history is rich: seizure disorders 35 %, headache 28 %, sleep disorders 22 %, depression 20 %, stroke / cerebrovascular 15 %, movement disorders 15 %, dementia 8 %, tumours 6 %.
  - The TUH sheet notes much less per session: seizure / epilepsy 9 %, stroke 4 %, substance use 4 %, and keyword mentions such as altered mental status 8 % or syncope 4 %.
  - TUAB's diagnosis columns are essentially empty (Section 1).
  - Rates are therefore comparable *within* a corpus, not across corpora.
- **Headache / migraine** is the most common non-seizure category in HEEDB (28 %).
  - **Sex:** about 1.4× more frequent in women (32 vs 23 %); the TUH mentions show the same sex ratio (TUAB 7.4 vs 2.9 %, TUH cohort 3.0 vs 1.3 %).
  - **Age:** in TUH it is a disease of the young (3–4 % of mentions under 50, < 1 % above 60); HEEDB's lifetime codes stay at 23–32 % across ages (Figure 3.3).
  - **In TUH it marks a normal EEG:** 10 % of TUAB and 33 % of TUH-cohort recordings with a headache mention are abnormal, against 46 % and 58 % without (Table 3.3).
  - **In HEEDB it has no such effect** (60 % vs 60 %), because the ICD code is lifetime history and co-occurs with other neurological disease: of HEEDB headache patients' recordings, 52 % also carry a seizure disorder, 38 % depression and 39 % a sleep disorder (Table 3.4).
- **Pathologies that raise the abnormal share most** (Eq. (3.2), Table 3.3):
  - stroke (TUH cohort 82 % vs 56 % abnormal, +26 pp);
  - anoxic injury (+24 pp) and altered mental status (+23 pp);
  - brain tumour (+14 pp TUH, +19 pp HEEDB) and CNS infection (+18 pp HEEDB);
  - dementia (+15 pp TUH, +15 pp HEEDB).
- **Pathologies that do not raise it.** Psychiatric and benign neurological categories go with an equal or *lower* abnormal share:
  - TUH cohort: depression −13 pp, anxiety −9, ADHD −14, MS −6, headache −25, syncope −24.
  - HEEDB: depression −1, MS −3, headache 0, ADHD −16.
  - Their patients are younger and their diagnoses leave no consistent EEG trace, so they are hard targets for an EEG classifier.
- **Sex differences beyond headache:**
  - substance use is twice as frequent in men in TUH (5.6 vs 2.9 %, TUAB 6.0 vs 2.5 %);
  - in HEEDB, depression (23 vs 17 %) and multiple sclerosis (1.9 vs 0.9 %) are more frequent in women;
  - stroke is slightly more frequent in men in both corpora (4.2 vs 3.7 %, 15.5 vs 13.7 %).
- **Comorbidity is the rule in HEEDB:** 43 % of the dementia patients' recordings also carry a seizure disorder, and 33 % stroke (Table 3.4). Per-pathology targets are therefore far from mutually exclusive, and a multi-label (not multi-class) formulation is the right one for the pathology heads.

Abnormality types (Tables 4.1–4.3, Figure 4.1).

- **Slowing dominates.**
  - TUH: diffuse slowing in 61 % and focal slowing in 44 % of abnormal reasons, epileptiform discharges in 34 %, suppression / low voltage in 20 %, seizures in 4 %.
  - HEEDB's structured findings name generalised slowing in 87 %, epileptiform or periodic discharges in 38 %, focal slowing in 23 % and a seizure in 17 % of abnormal recordings.
  - The higher HEEDB generalised-slowing and seizure rates partly reflect structured coding (every mention counts) against keyword rules on a short TUH reason text.
- **Types follow the pathologies as expected** (lift, Eq. (4.2)):
  - TUH: seizures are 4.4× over-represented and epileptiform discharges 1.3× in seizure / epilepsy patients; suppression / low voltage is 2.5× over-represented after anoxic injury; focal slowing 1.7× with brain tumours and 1.3× after stroke.
  - HEEDB shows the same focal-slowing lift for tumours (1.7×) and the epileptiform lift for seizure disorders (1.3×).
- **Rare TUH groups, wide error bars.** In the TUH cohort the rarer categories (MS, ADHD, schizophrenia, dementia) have under 100 positives each. Their prevalence and lifts carry wide uncertainty and should not be over-read.

What this means for the pathology model (E2P). E2P (TUH + HEEDB, status + 17 diagnosis groups, `dataset_maker/make_multitask_targets.py`) inherits these properties:

- **HEEDB dominates most groups.** It provides most positives for seizure, stroke, dementia, depression and all of headache, sleep, tumour and movement disorders.
- **TUH-only groups.** The psychiatric, substance and TBI groups come from TUH only (HEEDB unknown).
- **Undated labels.** HEEDB groups are lifetime diagnoses, so EEG-invisible history (e.g. a remote headache code) caps the reachable ROC-AUC.
- **Expected detectability** (Table 3.3): stroke, tumour, CNS infection, dementia and the seizure groups should be detectable above chance; depression, ADHD, MS and headache will be close to it.

6. The classes predicted by E2P, the pathology model

E2P (`/data/runs/heedb/submit_e2p_1gpu.sh` , docs/heedb_onboarding_plan.md) trains on the joint TUH + HEEDB cohort `tuh_heedb_v1` , on E2 / E2L's split, a **multi-label** classifier without an age prediction: 18 independent sigmoid outputs. Its targets are `multitask_targets_v2.npz` (`dataset_maker/make_multitask_targets.py`):

- `status_abnormal` : the EEG report's verdict (1 = abnormal);
- 13 NEDC diagnosis groups: `clinical_seizure` , `electrographic_seizure` , `epilepsy` , `stroke` , `tbi` , `substance` , `bipolar` , `anxiety` , `depression` , `schizophrenia` ,

`dementia` (incl. Alzheimer's), `ms` and `other` (Parkinson's, ADHD). TUH from the NEDC sheet; HEEDB from the ICD-10 based `dx_*` flags;
- 4 HEEDB ICD-10 groups: `headache_disorders` (incl. migraine), `sleep_disorders`, `intracranial_and_spinal_tumors`, `movement_and_cerebellar_disorders` (unknown for TUH).

A label is unknown, and left out of the loss and the metrics, when the recording has no report verdict (status), when its session is not in the NEDC sheet (TUH), when the patient has no ICD-10 record (HEEDB), or when the dataset does not code the group at all (blank rows below).

Table 6.1. Targets: known labels, positives and prevalence per dataset

All splits together. `known` = recordings whose label is known, `positive` = positives, `%` = prevalence, Eq. (3.1).

	known		positive		%	
	TUH	HEEDB	TUH	HEEDB	TUH	HEEDB
target						
status_abnormal	28854	36448	16591	21986	57.5	60.3
clinical_seizure	28554	65020	2228	23035	7.8	35.4
electrographic_seizure	28554	65020	378	5153	1.3	7.9
epilepsy	28554	65020	846	22931	3.0	35.3
stroke	28554	65020	1122	7363	3.9	11.3
tbi	28554	0	364	0	1.3	–
substance	28554	0	1214	0	4.3	–
bipolar	28554	0	321	0	1.1	–
anxiety	28554	0	122	0	0.4	–
depression	28554	65020	129	13024	0.5	20.0
schizophrenia	28554	0	92	0	0.3	–
dementia	28554	65020	83	4990	0.3	7.7
ms	28554	65020	66	934	0.2	1.4
other	28554	65020	39	3255	0.1	5.0
headache_disorders	0	65020	0	17954	–	27.6
sleep_disorders	0	65020	0	14356	–	22.1
intracranial_and_spinal_tumors	0	65020	0	4126	–	6.3
movement_and_cerebellar_disorders	0	65020	0	9477	–	14.6

Table 6.2. Positives per split and dataset

Positive recordings of each target in the train / validation / test splits (E2's split). TUH test is TUAB eval: the NEDC sheet hardly annotates diagnoses for TUAB sessions (Section 1), so the TUH test split has almost no pathology positives, and TUH pathology can only be judged on validation. HEEDB counts under 10 are blank.

	train		val		test	
	TUH	HEEDB	TUH	HEEDB	TUH	HEEDB
target						
status_abnormal	14790	17554	1676	2179	125	2253
clinical_seizure	2043	18369	185	2302	0	2364
electrographic_seizure	367	4120	11	538	0	495
epilepsy	800	18284	46	2292	0	2355
stroke	1028	5891	93	735	1	737
tbi	314	0	50	0	0	0
substance	1101	0	103	0	10	0
bipolar	283	0	31	0	7	0
anxiety	112	0	10	0	0	0
depression	121	10404	8	1272	0	1348
schizophrenia	83	0	9	0	0	0
dementia	72	3959	10	514	1	517
ms	59	744	7	91	0	99
other	27	2581	12	324	0	350
headache_disorders	0	14398	0	1775	0	1781
sleep_disorders	0	11543	0	1408	0	1405
intracranial_and_spinal_tumors	0	3329	0	383	0	414
movement_and_cerebellar_disorders	0	7561	0	893	0	1023

Table 6.3. Prevalence per split and dataset (%)

Eq. (3.1) per split and dataset, among the recordings whose label is known.

	train		val		test	
	TUH	HEEDB	TUH	HEEDB	TUH	HEEDB
target						
status_abnormal	58.0	60.5	54.7	59.3	45.6	59.7
clinical_seizure	8.1	35.4	6.2	35.5	0.0	35.5
electrographic_seizure	1.5	7.9	0.4	8.3	0.0	7.4
epilepsy	3.2	35.2	1.5	35.3	0.0	35.4
stroke	4.1	11.4	3.1	11.3	0.4	11.1
tbi	1.2	–	1.7	–	0.0	–
substance	4.4	–	3.4	–	3.7	–
bipolar	1.1	–	1.0	–	2.6	–
anxiety	0.4	–	0.3	–	0.0	–
depression	0.5	20.1	0.3	19.6	0.0	20.3
schizophrenia	0.3	–	0.3	–	0.0	–
dementia	0.3	7.6	0.3	7.9	0.4	7.8
ms	0.2	1.4	0.2	1.4	0.0	1.5
other	0.1	5.0	0.4	5.0	0.0	5.3
headache_disorders	–	27.8	–	27.4	–	26.8
sleep_disorders	–	22.2	–	21.7	–	21.1
intracranial_and_spinal_tumors	–	6.4	–	5.9	–	6.2
movement_and_cerebellar_disorders	–	14.6	–	13.8	–	15.4